

PURITY KARWITHA KIRIMI

AI PRODUCT ENGINEER | APPLIED AI | MACHINE LEARNING

+254 706 470 548 • purity.kirimi.k@gmail.com • Nairobi, Kenya
linkedin.com/in/purity-karwitha-9357a41b6 • github.com/pkirimi

POSITIONING

I build practical AI products and machine-learning systems that turn real problems into useful, measurable decision-support and workflow improvements.

PROFILE

AI Product Engineer with a clinical medicine background and hands-on experience building AI applications, machine-learning models and user-facing products. I work across problem framing, data preparation, model evaluation, LLM/API integration, product development and deployment. My approach is practical: understand the decision or workflow first, then use the simplest technology that can create useful value.

TECHNICAL CAPABILITIES

AI & LLM

LLM APIs • RAG • Prompt engineering • Embeddings • AI agents • NLP • Evaluation

Machine Learning

Python • Pandas • NumPy • scikit-learn • Classification • Feature analysis • Model evaluation

AI Application Engineering

Streamlit • FastAPI • Hugging Face • API integration • Git/GitHub • Deployment

Product

Problem framing • User journeys • Decision support • Workflow design • Responsible AI • Prototyping

EXPERIENCE

AI Engineering & Product Development | Independent / Project-Based

Kenya • AI engineering training + applied product work

- Built and shipped practical AI/ML projects spanning LLM applications, conversational AI, RAG, classification and customer analytics.
- Worked end-to-end from problem framing and data exploration to application development, API integration and deployment.
- Used AI as a product-building partner while maintaining human review of technical decisions, model behaviour and final outputs.

Clinical Medicine | Healthcare

Kenya • 6 years clinical experience

- Developed structured problem-solving, evidence-based decision-making and clear communication in complex real-world environments.
- Built strong domain understanding of healthcare workflows, patient needs and the importance of context, safety and appropriate escalation.

SELECTED PROJECTS

HerCycle

AI Product • Mobile • Women's Health

Built and shipped an AI-powered women's health product for African women, spanning multiple user journeys. Worked across product design, personalised AI interactions, mobile development, backend services and deployment.

MediAssist Kenya

Conversational AI • RAG • Healthcare

Built a clinical AI assistant exploring retrieval, embeddings, LLM generation and citation-supported responses. Deployed the project on Hugging Face and explored practical constraints such as context, latency and responsible decision support.

Search Intelligence

Machine Learning • Classification • FlyRank

Built a classification workflow on 30,000 rows of search/content intelligence data. Logistic Regression reached 0.946 recall and 0.702 F1, while ROC AUC of 0.515 exposed weak overall discrimination—an important model-evaluation finding.

Diabetes Risk Prediction
Classical ML • Model Comparison

Compared Logistic Regression and Random Forest on the Pima Indians Diabetes dataset. Random Forest achieved approximately 0.7468 test accuracy; the project focused on data quality, evaluation and model comparison.

Telco Customer Churn
Customer Analytics • Business ML

Framed customer churn as a business decision-support problem: identify patterns associated with customers who may leave and connect predictions to retention prioritisation rather than treating modelling as the end goal.

30K
SEARCH DATA ROWS
0.946
SEARCH ML RECALL
0.7468
DIABETES RF ACCURACY
LIVE
HER CYCLE PRODUCT

EDUCATION & TRAINING

AI Engineering Training — completed

Clinical Medicine — professional background

TARGET ROLES

AI Engineer • Applied AI Engineer • AI Product Engineer • ML Engineer • Machine Learning Engineer • AI Application Engineer • AI Solutions Engineer • Generative AI / LLM Engineer

LINKS

Portfolio: <https://pkirimi.github.io/Purity-Portfolio/>

GitHub: <https://github.com/pkirimi>

LinkedIn: <https://www.linkedin.com/in/purity-karwitha-9357a41b6>

MediAssist: <https://huggingface.co/spaces/Purityk/mediassist>